BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The respiratory system

The written notes for the respiratory lectures. Module 4 competencies 97 to 104, with eight worked questions at the end.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Before the competencies: what the system does

In your course notes: Module 4 packet, Chapter 1 introduction (page 4).

Dr. Rennie's video for this part: Respiratory system concept video, chapter Respiratory system overview, and Ventilation and gas exchange.

Respiration is four linked jobs. Two of them belong to this system, one belongs to the cardiovascular system, and the last one happens out in the tissues. The respiratory system and the cardiovascular system are so tightly linked that if either fails, oxygen never reaches the cells.

1. Pulmonary ventilation (breathing): air moves into the lungs (inspiration) and out of the lungs (expiration), like a bellows.
2. External respiration: oxygen crosses from the alveoli into the blood and carbon dioxide crosses from the blood into the alveoli, at the respiratory membrane.
3. Transport of gases: the cardiovascular system carries oxygen from the lungs to the tissues and carbon dioxide back. The heart and vessels are the delivery truck; the blood is the cargo.
4. Internal respiration: oxygen moves from the systemic capillaries into the tissue cells and carbon dioxide moves out of them.

- Cellular respiration is something else: the mitochondria using oxygen and making carbon dioxide to produce ATP. It is a job of every cell, not of the respiratory system.
- Moving air also gives you smell and speech.
- The structures in order, nose to alveolus: nose, nasal cavity, paranasal sinuses, pharynx, larynx, trachea, bronchi and their branches, lungs, alveoli. (Nice Nurses Practice Thorough Bronchial Lung Assessments.)
- The respiratory muscles, the diaphragm and its helpers, create the pressure changes that move the air. They are competency 103.

Competency 97. Upper and lower tracts, conducting and respiratory zones

In your course notes: Module 4 packet, Chapter 1, competency 97 (page 4).

Dr. Rennie's video for this part: Respiratory system concept video, chapter Respiratory system overview.

You should be able to: Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones, and assign the major airway structures to each.

The respiratory system moves air into and out of the body and brings that air close to the blood. Air follows

one continuous path from the nose to the air sacs of the lungs. Anatomists group that path two ways, and the two groupings use different dividing lines, so you have to learn both.

One breath, step by step

1. Air enters at the nares (nostrils) and passes the nasal vestibule, where coarse hairs catch large particles.
2. It moves through the nasal cavity, tumbling past the conchae, and is warmed and moistened.
3. It leaves through the choanae into the nasopharynx (air only), then the oropharynx and laryngopharynx (shared with food).
4. It turns forward into the larynx and passes the glottis, between the vocal folds.
5. It runs down the trachea to the carina, where the trachea forks.
6. It enters a main bronchus (one per lung), then a lobar bronchus (one per lobe), then a segmental bronchus (one per bronchopulmonary segment).
7. It passes through bronchioles, which have no cartilage, to a terminal bronchiole, the last part of the conducting zone.
8. It crosses into a respiratory bronchiole, the first airway of the respiratory zone, then an alveolar duct, an alveolar sac, and an alveolus, where gases cross into the blood.

Upper respiratory tract
: The nose, nasal cavity, paranasal sinuses, and pharynx. These sit in the head and neck.

Lower respiratory tract
: The larynx, trachea, bronchi and all their branches, and the lungs. The larynx is the first structure of the lower tract, so the line between the tracts falls where the pharynx meets the larynx. Some clinical sources put the larynx in the upper tract; these notes follow the anatomy textbook split.

Conducting zone
: Every passage that only carries air, from the nose down to the terminal bronchioles. Its walls are too thick for gases to cross. It warms, moistens, and filters the air on the way through.

Respiratory zone
: Where gas exchange happens: the respiratory bronchioles, alveolar ducts, alveolar sacs, and alveoli. The line between the zones is deep inside the lung, where a terminal bronchiole gives way to a respiratory bronchiole.

Structure	Tract	Zone
Nose and nasal cavity	Upper	Conducting
Paranasal sinuses	Upper	Air spaces off the nasal cavity, no gas exchange
Pharynx	Upper	Conducting
Larynx	Lower	Conducting
Trachea	Lower	Conducting
Main, lobar, and segmental bronchi	Lower	Conducting
Bronchioles and terminal bronchioles	Lower	Conducting
Respiratory bronchioles	Lower	Respiratory
Alveolar ducts and sacs	Lower	Respiratory
Alveoli	Lower	Respiratory

- Tract tells you where a structure is. Zone tells you what it does.
- Because the lines fall in different places, one structure can be lower tract and conducting zone at the same time. The trachea is the usual example.
- Every upper tract structure is conducting. The lower tract holds the rest of the conducting zone plus the whole respiratory zone.

- The conducting zone ends at the terminal bronchioles, not at the bronchi and not where the lung begins. Gas exchange itself is physiology; this course covers the structures.

Competency 98. Nose, nasal cavity, and pharynx

In your course notes: Module 4 packet, Chapter 1, competency 98 (page 6).

Dr. Rennie's video for this part: Respiratory system concept video, chapter Respiratory system overview.

You should be able to: Identify the parts of the nasal cavity, paranasal sinuses, and the three regions of the pharynx.

The nose is the main entrance for air. It warms, moistens, and filters the air, it houses the receptors for smell, and it acts as a resonating chamber for the voice. Bone forms the upper part of the external nose (nasal bones, frontal bone, maxillae) and hyaline cartilage forms the lower part, which is why the tip bends.

The external nose

Feature	What it is
Root	Where the nose meets the forehead, between the eyebrows (the glabella is the flat area between the brows)
Bridge	The upper, bony part
Dorsum nasi	The anterior margin, from the root to the apex
Apex	The tip
Alae	The flared sides of the nostrils
Nares	The nostrils, the external openings

- Framework: nasal bones and the frontal bone above, the maxillae at the sides, and plates of hyaline cartilage below (septal and alar cartilages and the lateral processes of the septal cartilage).
- The nose does five things: it is an airway, it warms and moistens the air, it filters and cleans it, it resonates for speech, and it houses the smell receptors.

Parts of the nasal cavity

Nasal septum
The midline wall that splits the cavity into right and left halves. Septal cartilage in front; the perpendicular plate of the ethmoid above the vomer behind.

Nasal vestibule
The area just inside each nostril, lined with skin, sebaceous glands, and coarse hairs that trap large particles.

Nasal conchae
Three scroll-shaped shelves on each lateral wall: superior, middle, and inferior. The superior and middle conchae are parts of the ethmoid bone. The inferior concha is a separate bone. They make the air tumble so more of it touches the warm, moist lining.

Nasal meatuses
The passages under each concha: superior, middle, and inferior meatus. A meatus is the space, not the shelf.

Hard palate
The bony front part of the floor, separating the nasal cavity from the mouth.

Soft palate
The muscular back part of the floor. It lifts to close off the nasopharynx during swallowing.

Choanae
The two back openings where the nasal cavity becomes the nasopharynx.

Two linings

Mucosa	Epithelium	Job
Olfactory mucosa	Olfactory epithelium, a small patch in the roof	Houses the smell receptors
Respiratory mucosa	Pseudostratified ciliated columnar with goblet cells	Secretes mucus that traps dust and microbes; the cilia sweep it back to the pharynx to be swallowed

- The vestibule is lined with stratified squamous epithelium, because it takes the mechanical wear at the entrance.
- The seromucous glands secrete lysozyme, an antibacterial enzyme, and the epithelial cells secrete defensins. The high water content of the mucus humidifies the air.
- A dense capillary bed close to the surface warms incoming air. It is also why nosebleeds (epistaxis) are common.
- Nerve endings in the lining trigger the sneeze reflex when dust or pollen irritates them.
- Cold air slows the cilia, so mucus stops moving back toward the throat and drips out the front. That is why your nose runs in cold weather. On the way out, water vapor condenses on the cooler lining, so the conchae reclaim heat and moisture during exhalation.

The paranasal sinuses

Air-filled spaces inside four skull bones around the nasal cavity, each named for its bone and lined with the same mucosa as the nasal cavity. Each drains through a small opening into the nasal cavity. They lighten the skull, help warm and moisten air, and add resonance to the voice.

Sinus	Bone	Where it sits	Drains into
Frontal sinus	Frontal	In the forehead, above the orbits	Middle meatus
Ethmoidal air cells	Ethmoid	Many small spaces between the nasal cavity and the orbits	Middle meatus (front and middle cells), superior meatus (back cells)
Sphenoidal sinus	Sphenoid	Deep in the skull, behind the nasal cavity	The sphenoethmoidal recess, above the superior concha
Maxillary sinus	Maxilla	In the cheek, below each orbit; the largest	Middle meatus

The pharynx, top to bottom

The pharynx (throat) is a funnel-shaped muscular tube from the base of the skull down to the level of the cricoid cartilage, where it continues as the esophagus. It connects the nasal cavity and mouth to the larynx and esophagus.

Region	Location	Carries	Epithelium and structures
Nasopharynx	Behind the nasal cavity, above the soft palate	Air only	Pseudostratified ciliated columnar. Pharyngeal tonsil (adenoid) on the back wall, auditory tube openings on the side walls
Oropharynx	Behind the oral cavity, soft palate to epiglottis	Air and food	Stratified squamous, which resists friction. Palatine tonsils in the side walls, lingual tonsil on the back of the tongue
Laryngopharynx	Behind the larynx, epiglottis to esophagus	Air and food	Stratified squamous. Where air turns forward into the larynx and food goes back into the esophagus

- When you swallow, the soft palate and uvula lift and seal the nasopharynx, so food does not enter the nose.
- The epithelium changes with the job. Where only air passes, the lining is ciliated and moves mucus. Where food also passes, it is stratified squamous and stands up to rubbing.

Competency 99. Larynx

In your course notes: Module 4 packet, Chapter 1, competency 99 (page 8).

Dr. Rennie's video for this part: Respiratory system concept video, chapter Larynx and voice passage.

You should be able to: Identify the cartilages and folds of the larynx, including the thyroid and cricoid cartilages, epiglottis, arytenoids, vocal cords, and glottis.

The larynx (voice box) connects the laryngopharynx to the trachea. It sits in the front of the neck at about C3 to C6 and hangs from the hyoid bone, which is above it and not part of it. It has three jobs: hold the airway open, keep food and drink out of the lower airway, and produce the voice.

The nine cartilages

Cartilage	Number	Cartilage type	What it is
Thyroid cartilage	Single	Hyaline	The large shield. Its front ridge is the laryngeal prominence (Adam's apple). Open at the back
Cricoid cartilage	Single	Hyaline	The signet ring just below the thyroid cartilage, narrow in front and wide behind. The only complete ring of cartilage in the airway. The trachea hangs from its lower edge
Epiglottis	Single	Elastic	The spoon-shaped flap attached inside the thyroid cartilage. It tips down over the laryngeal inlet during a swallow
Arytenoid cartilages	Paired	Mostly hyaline	Sit on top of the back of the cricoid. They anchor the vocal folds and pivot to move them, so they are the most important paired cartilages
Corniculate cartilages	Paired	Elastic	Small cartilages on top of the arytenoids
Cuneiform cartilages	Paired	Elastic	Small cartilages in the fold running from each arytenoid up to the epiglottis

- The cricothyroid ligament fills the gap between the thyroid and cricoid cartilages in front. It lies just under the skin, below the vocal folds, with no cartilage over it, which is why it is the landmark for an emergency airway when the larynx above is blocked.
- The epiglottis is elastic because it bends over the inlet with every swallow and springs back. The thyroid, cricoid, and arytenoid cartilages are hyaline (only the tips of the arytenoids are elastic).

The folds and the glottis

Vocal folds

The true vocal cords, the lower pair. Each contains a vocal ligament, an elastic band running from an arytenoid cartilage behind to the thyroid cartilage in front. Very few blood vessels, so they look pearly white. They vibrate as air passes between them, and that vibration is sound.

Vestibular folds

The false vocal cords, the upper pair. They help close the airway during swallowing and play no part in sound.

Glottis

The vocal folds together with the opening between them (the rima glottidis), the narrowest point air passes in the larynx. Books draw the edges a little differently: some use the name for the opening alone, and OpenStax includes the vestibular folds.

Laryngeal inlet

The opening into the larynx from the laryngopharynx, which the epiglottis covers during a swallow.

- When the arytenoids pivot, the vocal folds swing apart to widen the glottis for breathing or come together to close it.
- The lining changes at the vocal folds. Above them, where food can rub, stratified squamous. Below them, pseudostratified ciliated columnar, with cilia sweeping mucus up toward the pharynx.
- The vocal folds make the sound, but the pharynx, oral cavity, and sinuses amplify it like the body of an instrument, and the lips, tongue, and soft palate shape it into words.

- The larynx also works as a sphincter. When you strain, lift something heavy, or empty your bowels, the vocal folds close and hold air in the chest (the Valsalva maneuver), which stiffens the trunk and raises abdominal pressure.
- The laryngeal prominence grows larger in males at puberty under testosterone, which is why the voice drops.

Competency 100. Trachea and bronchial tree

In your course notes: Module 4 packet, Chapter 1, competency 100 (page 10).

Dr. Rennie's video for this part: Respiratory system concept video, chapter Trachea and breathing tubes.

You should be able to: Trace air through the trachea and bronchial tree in order from the trachea and carina down to the terminal bronchioles.

Below the larynx the airway becomes a branching tree. The tubes get narrower and more numerous at every step, and every one of them down to the terminal bronchioles belongs to the conducting zone.

From the trachea to a terminal bronchiole, in order

1. Trachea: the windpipe, from the lower edge of the cricoid cartilage into the mediastinum, about 10 to 12 cm long. Sixteen to twenty C-shaped rings of hyaline cartilage hold it open. The open side of each C faces back toward the esophagus, and the trachealis muscle bridges the gap, so the esophagus can bulge forward when you swallow.
2. Carina: the internal ridge formed by the lowest tracheal cartilage at the fork, about at the sternal angle (T4 to T5). Its lining is very sensitive; touching it sets off a strong cough.
3. Main (primary) bronchi: one enters each lung at the hilum. The right main bronchus is wider, shorter, and more vertical than the left, so an inhaled object usually lands in the right lung.
4. Lobar (secondary) bronchi: one to each lobe, three on the right and two on the left.
5. Segmental (tertiary) bronchi: one to each bronchopulmonary segment.
6. Bronchioles: tubes less than 1 mm across, with smooth muscle and no cartilage.
7. Terminal bronchioles: the last and smallest conducting tubes. The next tubes, the respiratory bronchioles, begin the respiratory zone.

The four layers of the tracheal wall

Layer	What is in it
Mucosa	Pseudostratified ciliated columnar epithelium with goblet cells on a lamina propria rich in elastic fibers. The cilia move mucus up toward the pharynx
Submucosa	Connective tissue with seromucous glands that produce sheets of mucus
Cartilaginous layer	The 16 to 20 C-shaped rings of hyaline cartilage, open at the back, bridged by the trachealis muscle. The last cartilage forms the carina
Adventitia	The outer connective tissue that ties the trachea to its neighbors

- The elastic fibers let the trachea stretch and recoil with each breath.
- When the trachealis contracts it narrows the trachea, which speeds the air during a cough and helps blast secretions out.

How the wall changes down the tree

Level	Cartilage	Epithelium	Smooth muscle
Trachea	C-shaped hyaline rings	Pseudostratified ciliated columnar with goblet cells	Trachealis across the back of each ring
Main bronchi	C-shaped rings	Pseudostratified ciliated columnar with goblet cells	A thin layer
Lobar and segmental bronchi	Irregular plates	Pseudostratified ciliated columnar, fewer goblet cells in smaller branches	A more complete layer
Bronchioles	None	Simple columnar to simple cuboidal	A complete layer, much of the wall
Terminal bronchioles	None	Simple cuboidal	A complete layer

- Going down the tree, cartilage shrinks and smooth muscle takes over. Rings, then plates, then none.
- Cartilage is already gone at the bronchioles, not just at the terminal bronchioles.
- Without cartilage to hold them open, bronchioles narrow a great deal when their smooth muscle contracts. That is the structural reason asthma is so dangerous.

Competency 101. Lungs and pleurae

In your course notes: Module 4 packet, Chapter 1, competency 101 (page 11).

Dr. Rennie's video for this part: Respiratory system concept video, chapter Lungs position and diaphragm, and Lung sections and blood flow.

You should be able to: Identify the lobes, fissures, and surfaces of each lung and the parietal and visceral pleurae with the pleural cavity.

The two lungs fill most of the thoracic cavity, one on each side of the mediastinum (the central compartment holding the heart, great vessels, trachea, and esophagus), with the diaphragm beneath them. Each lung is cone-shaped, soft, and spongy. The two are not the same size: the heart tilts left and takes room, so the left lung is smaller; the liver pushes the right dome of the diaphragm up, so the right lung is shorter and wider.

Surfaces and landmarks

Apex
The narrow top of the lung, rising just above the clavicle into the root of the neck.

Base
The broad, curved bottom surface resting on the diaphragm.

Costal surface
The large outer surface against the ribs.

Mediastinal surface
The medial surface facing the mediastinum. The hilum is on it.

Hilum
The indented area on the mediastinal surface where the bronchus, blood vessels, nerves, and lymphatic vessels enter and leave.

Root of the lung
The bundle passing through the hilum: main bronchus, pulmonary artery, pulmonary veins, bronchial vessels, nerves, and lymphatics.

Cardiac notch
The indentation in the front edge of the left lung that makes room for the heart.

Lingula
The tongue-shaped part of the left superior lobe just below the cardiac notch. It sits where the middle lobe sits on the right.

- In the root, the pulmonary artery carries oxygen-poor blood into the lung and the pulmonary veins carry oxygen-rich blood back to the heart.
- Telling right from left at the hilum on the cadaver: in the left hilum the pulmonary artery is the highest structure; in the right hilum a bronchus sits at or above the artery. In both, the pulmonary veins are lowest.

Lobes and fissures

Lung	Lobes	Fissures	Notes
Right lung	Three: superior, middle, inferior	Oblique fissure (separates the inferior lobe) and horizontal fissure (separates superior from middle)	Larger; shorter and wider because the liver raises the right diaphragm. Ten bronchopulmonary segments
Left lung	Two: superior, inferior	Oblique fissure only	Smaller, with the cardiac notch and the lingula. About 8 to 10 segments

- Each lobe is divided into bronchopulmonary segments, each served by its own segmental bronchus, artery, and vein. Because each segment is its own unit, a surgeon can remove a diseased segment without damaging the segments around it, like taking one section out of an orange.
- Right has three, left has two. The left gave up its middle lobe to make room for the heart, and the lingula is what is left in its place.

Two blood supplies to the lung

Circulation	Pressure and volume	Job
Pulmonary circulation (pulmonary arteries and veins)	Low pressure, high volume	Gas exchange at the alveoli
Bronchial circulation (bronchial arteries off the aorta)	High pressure, low volume	Oxygen-rich systemic blood to feed the lung tissue itself

The lungs are supplied by parasympathetic and sympathetic motor fibers and by visceral sensory fibers. The sympathetic fibers widen the airways; the parasympathetic fibers narrow them.

The pleurae

Parietal pleura
The serous membrane lining the inside of the thoracic wall, the top of the diaphragm, and the sides of the mediastinum.

Visceral pleura
The serous membrane clinging to the lung surface and dipping into the fissures. The two layers are one continuous sheet that turns back on itself at the hilum.

Pleural cavity
The very thin space between the two layers, holding only a film of pleural fluid. There is a right and a left pleural cavity, kept apart by the mediastinum.

Costodiaphragmatic recess
The low pocket where the parietal pleura turns from the ribs onto the diaphragm. The lowest part of the pleural cavity, where extra fluid collects.

- The fluid does two jobs: it lets the lung glide against the chest wall, and it holds the two layers together so the lung follows the chest wall as it expands.
- Because the two pleural cavities are separate, air or fluid in one does not collapse the other lung.

Competency 102. Respiratory zone and alveolar structure

In your course notes: Module 4 packet, Chapter 1, competency 102 (page 13).

Dr. Rennie's video for this part: Respiratory system concept video, chapter Alveoli and gas exchange.

You should be able to: Describe the structure of an alveolus and identify the type I and type II alveolar cells, alveolar macrophages, respiratory membrane, and pulmonary capillaries.

Past the terminal bronchioles the airway walls thin into tiny air sacs. The respiratory zone is the bulk of the lung tissue, and at low power it looks lacy because it is mostly air spaces separated by thin walls.

Respiratory bronchioles
The first airways of the respiratory zone, with a few alveoli budding from their walls.

Alveolar ducts
Passages whose walls are lined entirely with alveoli.

Alveolar sacs

Clusters of alveoli opening into a shared space, like a bunch of grapes.

Alveoli

The single tiny air sacs where gas exchange happens. The two lungs hold hundreds of millions of them.

Alveolar pores

Small openings connecting neighboring alveoli.

The alveolar wall and the respiratory membrane

Structure	What it is	Job
Type I alveolar cells	Very thin, flat simple squamous cells covering nearly all of the alveolar surface	Gases cross the wall through them
Type II alveolar cells	Rounder, cuboidal cells scattered among the type I cells	Secrete surfactant, which lowers surface tension and keeps the alveoli from collapsing; also secrete antimicrobial proteins
Alveolar macrophages	The dust cells, moving freely over the inner surface of the alveoli	Engulf dust, microbes, and debris that got past the conducting zone
Pulmonary capillaries	A dense net wrapping each alveolus, fed by branches of the pulmonary arteries and drained by the pulmonary veins	Bring blood within a membrane's thickness of the air
Respiratory membrane	The barrier between alveolar air and blood where a capillary presses on an alveolus	Three layers, air to blood: the type I alveolar cell, the fused basement membranes of the alveolus and capillary, and the capillary endothelial cell

- Type I is thin, for exchange. Type II makes surfactant. Macrophages clean up.
- Elastic fibers surround each alveolus and let the lung stretch on the way in and recoil on the way out.
- The membrane is extremely thin and the alveoli are extremely many, so the lungs have a huge surface for gas exchange. That surface is fragile, which is the story of emphysema.

Competency 103. Thoracic wall and diaphragm

In your course notes: Module 4 packet, Chapter 1, competency 103 (page 14).

Dr. Rennie's video for this part: Respiratory system concept video, chapter Breathing muscles and final tips.

You should be able to: Identify the diaphragm, the intercostal muscles, and the three structures passing through the diaphragm at T8, T10, and T12.

The lungs sit inside the thoracic wall: ribs and costal cartilages, the sternum in front, the thoracic vertebrae behind, muscle between the ribs, and the diaphragm closing the bottom. Breathing works by changing the size of this cage. The lungs follow the wall because the pleural fluid holds them to it.

The diaphragm

- A dome-shaped sheet of skeletal muscle forming the floor of the thoracic cavity and separating it from the abdominal cavity. Its fibers attach around the lower rib cage, the sternum, and the lumbar vertebrae and pull toward the central tendon, a flat sheet at the top of the dome.
- The right dome sits higher because the liver is under it.
- When it contracts it flattens and moves down, making the thoracic cavity taller. It is the main muscle of breathing.
- Supplied by the right and left phrenic nerves, from spinal nerves C3, C4, and C5. On the cadaver each phrenic nerve runs down the side of the pericardium, in front of the root of the lung. C3, 4, and 5 keep the diaphragm alive.

The intercostal muscles and the accessory muscles

Muscle	Fiber direction or shape	Role in breathing	Nerve
Diaphragm	Dome, fibers rising to the central tendon	Main muscle of inspiration; contracts, flattens, moves down	Phrenic nerves (C3 to C5)
External intercostals	Downward and forward, like hands sliding into front pockets	Lift the ribs in inspiration, quiet and forced	Intercostal nerves
Internal intercostals	At right angles to the externals, downward and backward	Pull the ribs down in forced expiration (a hard cough, blowing out candles)	Intercostal nerves

- Quiet expiration needs no muscle effort; the lungs and chest wall recoil on their own.
- Harder inspiration recruits the scalenes, sternocleidomastoid, and pectoralis minor. Forced expiration uses the internal intercostals and the abdominal muscles.

Three structures through the diaphragm

Vertebral level	Opening	Main structure	Also passes through
T8	Caval opening, in the central tendon	Inferior vena cava	
T10	Esophageal hiatus, in the muscle	Esophagus	Vagus nerves
T12	Aortic hiatus, behind the diaphragm between the crura	Aorta	Thoracic duct

- I-8, E-10, A-12. Inferior vena cava at T8, Esophagus at T10, Aorta at T12. These are vertebral levels, listed top to bottom. (I Eat At 8, 10, 12 o'clock.)
- Above the diaphragm the vessel is the thoracic aorta; below it, the abdominal aorta. The crura are the muscular legs that anchor the diaphragm to the lumbar vertebrae.

Vessels you will find in the upper thorax (lab)

Structure	Artery or vein	Where it is
Brachiocephalic trunk	Artery	First branch of the aortic arch, right side only; splits into the right subclavian and right common carotid
Common carotid arteries	Artery	Up the neck to the head; the right from the brachiocephalic trunk, the left straight off the arch
Subclavian arteries	Artery	Out to the upper limbs; the right from the brachiocephalic trunk, the left straight off the arch
Internal jugular veins	Vein	Drain the head, down the neck beside the common carotids
Subclavian veins	Vein	Drain the upper limbs
Brachiocephalic veins	Vein	Each forms where an internal jugular joins a subclavian vein; the two join to form the superior vena cava
Pulmonary arteries	Artery	From the heart into each lung at the hilum
Pulmonary veins	Vein	From each lung back to the heart

For each one, say the side and whether it is an artery or a vein before you name it.

Competency 104. Respiratory disorders by structure affected

In your course notes: Module 4 packet, Chapter 1, competency 104 (page 16).

You should be able to: Name common respiratory disorders and state which airway, alveolar, or pleural structure each one affects.

Most respiratory disorders do one of three things. Some narrow or block the airways. Some damage or fill the alveoli, so the exchange surface is lost. Some affect the pleurae and pleural cavity, so the lung cannot stay against the chest wall. Learn the structure first, then what happens to it.

Disorder	Structure affected	What changes in that structure
Rhinitis	Nasal mucosa	The lining is inflamed, swollen, and makes extra mucus (allergies or infection)
Sinusitis	Paranasal sinuses	The sinus lining is inflamed, often when the openings into the nasal cavity are blocked
Laryngitis	Larynx, vocal folds	The vocal folds are inflamed and swollen, cannot vibrate normally, and the voice turns hoarse
Asthma	Bronchi and bronchioles	The lining is inflamed, smooth muscle tightens, extra mucus collects, and the airways narrow. Bronchioles have no cartilage to hold them open
Chronic bronchitis	Bronchi	The lining stays inflamed and makes too much mucus, which blocks the airways
Cystic fibrosis	Bronchi and bronchioles	An inherited disorder; the mucus is thick and sticky, the cilia cannot clear it, and it clogs the airways
Emphysema	Alveoli	Alveolar walls break down and many small alveoli merge into fewer, larger spaces. The exchange surface shrinks and the lung loses elastic recoil. The change is permanent
COPD	Airways and alveoli	Chronic obstructive pulmonary disease, a long-term blockage of airflow; mainly emphysema and chronic bronchitis together
Pneumonia	Alveoli	An infection inflames the alveoli and fills them with fluid. The walls are not destroyed
Tuberculosis	Lung tissue	A bacterial infection that forms small nodules (tubercles) and scar tissue
Infant respiratory distress syndrome	Alveoli (type II cells)	A premature newborn's type II cells do not yet make enough surfactant, so the alveoli collapse with each breath
Lung cancer	Lung tissue	A malignant tumor, usually starting in the epithelium lining the airways or alveoli; strongly linked to smoking
Pleurisy	Pleurae	The pleural layers are inflamed and rough, so they rub with each breath and cause sharp pain
Pleural effusion	Pleural cavity	Extra fluid collects in the pleural cavity and presses on the lung; in an upright person it pools in the costodiaphragmatic recess
Pneumothorax	Pleural cavity	Air enters the pleural cavity, the layers separate, and the lung on that side collapses

What this means for patients

- Asthma narrows the bronchioles so much because they have no cartilage to hold them open.
- Emphysema and pneumonia both take away working alveoli, but only emphysema destroys the walls. Pneumonia fills the sacs; emphysema erases them.

- A pneumothorax collapses one lung, not both, because each lung has its own sealed pleural cavity. The same thin, sealed space that lets the lung glide is what makes it vulnerable.
- Because the right main bronchus is wider, shorter, and more vertical, an inhaled object usually ends up in the right lung.
- The cricothyroid ligament sits just under the skin below the vocal folds with no cartilage over it, so it is the landmark for an emergency airway when the larynx above is blocked.

Worked questions

Eight questions in the style of the lecture exam, one for each competency. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 97

Which structure is the last part of the conducting zone?

A. Segmental bronchus

B. Terminal bronchiole

C. Respiratory bronchiole

D. Alveolar duct

Answer: B. The conducting zone runs from the nose down to the terminal bronchioles. They are the last tubes that only carry air.

Why the others are wrong

- **A** A segmental bronchus is in the conducting zone, but the tree keeps branching past it into bronchioles, so it is not the last part.
- **C** The respiratory bronchiole is the first airway of the respiratory zone. It has alveoli in its wall, so gas exchange starts there.
- **D** Alveolar ducts are lined entirely with alveoli and sit deep in the respiratory zone.

QUESTION 2 · COMPETENCY 98

Which region of the pharynx is lined with pseudostratified ciliated columnar epithelium?

A. Nasopharynx

B. Oropharynx

C. Laryngopharynx

D. All three regions

Answer: A. The nasopharynx carries only air, so it keeps the ciliated, mucus-moving lining of the nasal cavity.

Why the others are wrong

- **B** The oropharynx carries food as well as air, so it is lined with stratified squamous epithelium that resists rubbing.
- **C** The laryngopharynx also carries food and is stratified squamous for the same reason.
- **D** Only the nasopharynx is ciliated. The lining changes where food starts to pass.

QUESTION 3 · COMPETENCY 99

Which laryngeal cartilage is the only complete ring of cartilage in the airway?

A. Thyroid cartilage

B. Epiglottis

C. Cricoid cartilage

D. Arytenoid cartilage

Answer: C. The cricoid is shaped like a signet ring, narrow in front and wide behind, and it goes all the way around. The trachea hangs from its lower edge.

Why the others are wrong

- **A** The thyroid cartilage is the biggest, but it is a shield that is open at the back, not a ring.
- **B** The epiglottis is a single elastic flap, not a ring.
- **D** The arytenoids are small paired cartilages sitting on the back of the cricoid. They anchor the vocal folds.

QUESTION 4 · COMPETENCY 100

An inhaled peanut is most likely to lodge in the right lung because the right main bronchus is

A. narrower, longer, and more horizontal than the left

B. wider, shorter, and more vertical than the left

C. lined with more cilia than the left

D. the only main bronchus with C-shaped cartilage

Answer: B. The right main bronchus is wider, shorter, and more vertical, so an object falling down the trachea tends to drop straight into it.

Why the others are wrong

- **A** That describes the left main bronchus, which angles more sharply to get around the heart.
- **C** Both main bronchi have the same ciliated lining. Cilia move mucus; they do not steer objects.
- **D** Both main bronchi have C-shaped cartilage rings. The difference is the angle and width, not the cartilage.

QUESTION 5 · COMPETENCY 101

A student labels a horizontal fissure on the left lung. What is wrong with the drawing?

A. Nothing; both lungs have a horizontal fissure

B. The left lung has only an oblique fissure

C. The horizontal fissure belongs on the base, not the front

D. Fissures are only visible on the right lung

Answer: B. The left lung has two lobes separated by one oblique fissure. The horizontal fissure separates the superior and middle lobes of the right lung, and the left lung has no middle lobe.

Why the others are wrong

- **A** Only the right lung has three lobes and therefore two fissures.
- **C** The horizontal fissure runs across the front and side of the right lung at about the level of the fourth rib; it is not on the base.
- **D** The left lung has a clearly visible oblique fissure. It is the horizontal fissure it lacks.

QUESTION 6 · COMPETENCY 102

Which cell secretes surfactant?

A. Type I alveolar cell

B. Type II alveolar cell

C. Alveolar macrophage

D. Capillary endothelial cell

Answer: B. Type II alveolar cells are the cuboidal cells scattered in the alveolar wall. They make surfactant, which lowers surface tension so the alveoli do not collapse.

Why the others are wrong

- **A** Type I cells are the thin, flat cells that make up most of the wall. Gases cross through them, but they do not make surfactant.
- **C** Alveolar macrophages are the dust cells. They engulf debris; they do not secrete surfactant.
- **D** The capillary endothelium is the blood-side layer of the respiratory membrane. It belongs to the vessel, not the alveolus.

QUESTION 7 · COMPETENCY 103

The esophagus passes through the diaphragm at the level of which vertebra?

A. T8

B. T10

C. T12

D. L1

Answer: B. The esophageal hiatus is at T10. I-8, E-10, A-12: the inferior vena cava at T8, the esophagus at T10, the aorta at T12.

Why the others are wrong

- **A** T8 is the caval opening in the central tendon, for the inferior vena cava.
- **C** T12 is the aortic hiatus, behind the diaphragm between the crura, for the aorta and thoracic duct.
- **D** Nothing on this list passes the diaphragm at L1. The celiac trunk and superior mesenteric artery branch off the aorta near that level, below the diaphragm.

QUESTION 8 · COMPETENCY 104

Which disorder is air in the pleural cavity with collapse of the lung on that side?

A. Pleural effusion

B. Pneumonia

C. Pneumothorax

D. Emphysema

Answer: C. Pneumothorax is air between the parietal and visceral pleurae. The seal breaks, the lung pulls away from the chest wall, and that lung collapses. Only one side is affected because each lung has its own pleural cavity.

Why the others are wrong

- **A** Pleural effusion is fluid in the pleural cavity, not air. It pools in the costodiaphragmatic recess and presses on the lung.
- **B** Pneumonia is an infection inside the alveoli, which fill with fluid. The pleural cavity is not involved.
- **D** Emphysema destroys alveolar walls inside the lung. It is not air in the pleural space.